Daily Practice AP Precalculus Unit 1

Topics: Rates of Change and Concavity (Change in Tandem, Average Rate of Change, Concavity, Linear vs. Quadratic Functions)

Section 1: Multiple Choice Questions

No calculator is allowed for these questions.

1. What is the average rate of change of $f(x) = x^2 - 4x + 5$ on the closed interval $[-1, 3]$?

(A) -8

(B) -2

(C) 2

(D) 4

2. The temperature $T(h)$ of a liquid, in degrees Celsius, is modeled by a function where h is measured in hours. What is the best interpretation of an average rate of change of -3 on the closed interval $[1, 4]$?

(A) The temperature decreases by 3 degrees between $h = 1$ and $h = 4$.

(B) The temperature is -3 degrees Celsius at $h = 4$.

(C) On average, the temperature decreases by 3 degrees Celsius per hour between $h = 1$ and $h = 4$.

(D) The temperature decreases at a rate of 3 degrees Celsius per hour at $h = 1$.

3. Which of the following describes a function whose average rate of change over equal-length intervals is increasing and positive?

(A) Increasing and concave down

(B) Increasing and concave up

(C) Decreasing and concave down

(D) Decreasing and concave up

4. The graph of a function f is shown to be decreasing and concave up on the open interval (a, b). Which of the following could be a table of values for f on this interval?

(A) $f(1) = 10$, $f(2) = 8$, $f(3) = 6$

(B) $f(1) = 10,\ f(2) = 5,\ f(3) = 2$

(C) $f(1) = 10,\ f(2) = 7,\ f(3) = 2$

(D) $f(1) = 2,\ f(2) = 5,\ f(3) = 10$

5. Let g be a function whose average rate of change over equal-length intervals changes at a constant, non-zero rate. Which of the following must be true?

(A) g is a linear function.

(B) g is a quadratic function.

(C) g has a constant average rate of change.

(D) The graph of g has no concavity.

6. Selected values of a quadratic function $h(x)$ are given in the table below.

x	$h(x)$
1	4
3	10
5	20
7	k

What is the value of k?

(A) 30

(B) 32

(C) 34

(D) 36

7. If $f(x)$ is a function such that its graph is concave down on the interval $(0, 5)$, which of the following **MUST** be true about the average rate of change of f over equal-length subintervals of $(0, 5)$?

(A) It is always negative.

(B) It is always positive.

(C) It is increasing.

(D) It is decreasing.

8. For a linear function $L(x) = mx + b$, which of the following represents the geometric meaning of the average rate of change between any two distinct points on its graph?

(A) The length of the line segment connecting the two points.

(B) The slope of the secant line, which is equal to the constant m.

(C) The area of the triangle formed by the points and the axes.

(D) The difference between the y-intercept and the x-intercept.

Section 2: Free Response Questions

Show all your work clearly. Do not use calculus methods (e.g., derivatives) to justify your answers.

FRQ 1. Let f be the function defined by $f(x) = -x^2 + 6x - 5$.

(a) Calculate the average rate of change of f on the closed interval $[0, 4]$. Show the setup for your calculation.

(b) Using average rates of change over the equal-length intervals $[0, 2]$ and $[2, 4]$, determine if the graph of f is concave up or concave down. Justify your answer.

(c) A linear function $g(x)$ passes through the points $(0, f(0))$ and $(4, f(4))$. Write the equation for $g(x)$.

FRQ 2. A continuous function $W(t)$ models the amount of water in a tank, measured in liters, at time t, measured in minutes. Selected values of $W(t)$ are given in the table below.

t	$W(t)$
0	12
2	20
4	32
6	48

(a) Calculate the average rate of change of $W(t)$ over the interval $[0, 4]$. Include units in your final answer and interpret its meaning in the context of the problem.

(b) Is the function $W(t)$ linear, quadratic, or neither? Justify your reasoning using successive differences.

(c) Based on the table, describe the concavity of $W(t)$ on the interval $[0, 6]$. Justify your answer.